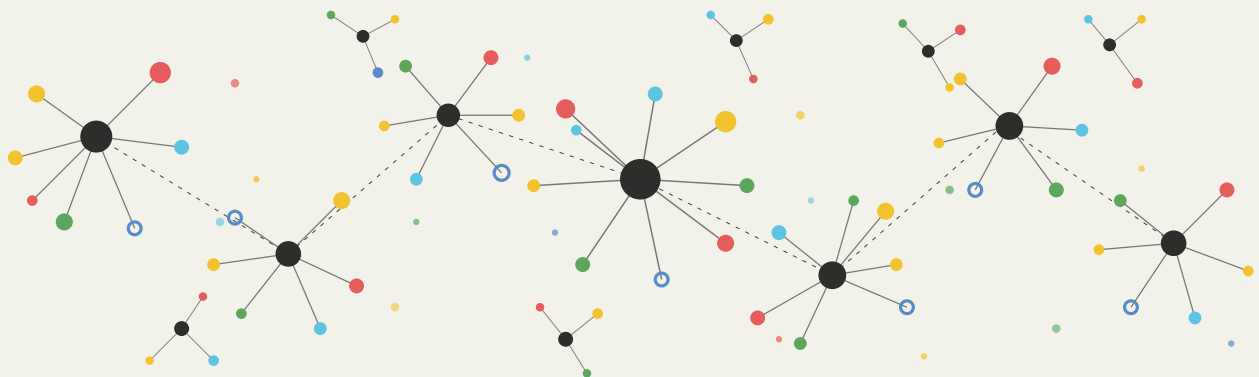


Growth Lab Data Visualization Spec



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Foundations — type & ink

Every chart is built from two type families and a small set of ink colors. Resist adding a third family or introducing new neutrals; reach for the existing tokens first.

SERIF — CHART TITLES & SOURCES

Source Serif 4

Weights used: 400, 500; italic for sources

Used for the chart title (14 px, weight 500) and the chart source line (12 px italic). Nothing else inside the figure block uses serif.

SANS — EVERYTHING ELSE

Inter

Weights used: 400, 500, 600

Subtitles, axis labels, axis ticks, series labels, annotations, and legends. Always set `font-variant-numeric: tabular-nums` on numeric ticks and any numeric tabular data so digits align.

CHART INK PALETTE

	--chart-title	#1A1714	Chart title, axis line, strong emphasis
	--chart-axis-label	#2C2823	Axis labels, tick labels, annotations
	--chart-subtitle	#4F4A42	Subtitles, secondary captions
	--chart-gridline	#D8D4CC	Horizontal/vertical gridlines inside the plot
	--chart-accent	#1A5A8E	Figure label, primary single-series default

Chart color palettes

CATEGORICAL — MAIN PALETTE

Seven categorical hues, each available in three tones: **light**, **main**, and **dark**. The main tone is the default fill for every mark — lines, bars, treemap. The dark tone is reserved for scatter-circle strokes **and all text associated with the colors** (direct labels, legend entries, callouts, annotations). The light tone is used for backgrounds, faded states, highlight bands, and the lightest steps of sequential ramps. Tokens follow the pattern — `-c-N-light` / `--c-N` / `---c-N-dark`.

	c-1	#B5D5EA · #2F87C8 · #1A5A8E	Primary — single-series default
	c-2	#E89C9C · #CC4948 · #8A2C2B	Contrast / lead-finding red
	c-3	#92D6BF · #2AA584 · #1A6B53	Third series
	c-4	#B5A0CC · #7554A3 · #4A3470	Fourth series
	c-5	#F4BC8A · #EA822D · #A8580F	Fifth series
	c-6	#E6E2A8 · #CDC86B · #8A8638	Sixth series
	c-muted	#CDD2D9 · #AFB5BE · #5F6773	De-emphasis — “everyone else”

Reading the tokens: `--c-1-light` (#B5D5EA) for a faint highlight wash; `--c-1` (#2F87C8) for the line, bar, or treemap tile; `---c-1-dark` (#1A5A8E) only for the stroke of a scatter circle. The same pattern repeats for every hue.

SEQUENTIAL — SINGLE-HUE RAMPS

Used on choropleth maps and any ordered encoding from low to high. Darker = higher value. Pick the ramp whose hue best matches the variable's meaning or the surrounding chart palette. **The examples below show five steps, but the number of steps should be adjusted based on the intended data representation** — three for a coarse classification, seven or more for a finer gradient.

c-1 · blue



#E5F0F9 #B5D5EA #6FA5CE #2F87C8 #1A5A8E

c-2 · red



#F4D5D5 #E89C9C #DC6F6E #CC4948 #8A2C2B

c-3 · teal



#D5EFE7 #92D6BF #5BC0A0 #2AA584 #1A6B53

c-4 · purple



#E5DDF0#B5A0CC#9276BA#7554A3#4A3470

c-5 · orange



#FBE5D5#F4BC8A#EE9A52#EA822D#A8580F

c-6 · yellow

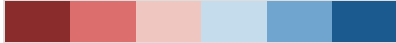


#FBF8DC#E6E2A8#DCD68E#CDC86B#8A8638

Diverging palettes

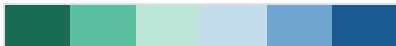
Used where data has a meaningful midpoint or reference point (gains vs. losses, above vs. below average). The boundary between the two hues marks the midpoint. Choose the pair whose semantics fit the variable. **The examples below show six steps, but the number of steps should be adjusted based on the intended data representation** — fewer for a coarse signal of direction, more for a finer gradient.

c-2 ↔ c-1 · red to blue



#8A2C2B #DC6F6E #EFC7C0 #C5DCEC #6FA5CE #1A5A8E

c-3 ↔ c-1 · teal to blue



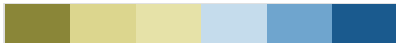
#1A6B53 #5BC0A0 #BDE5D8 #C5DCEC #6FA5CE #1A5A8E

c-5 ↔ c-1 · orange to blue



#A8580F #EE9A52 #F4BC8A #C5DCEC #6FA5CE #1A5A8E

c-6 ↔ c-1 · yellow to blue



#8A8638 #DCD68E #E6E2A8 #C5DCEC #6FA5CE #1A5A8E

Chart typography

FIGURE LABEL

FIGURE 4

Top of every figure block. Accent blue, uppercase. Always numbered sequentially.

CLASS	.figlbl
FAMILY	Inter
SIZE	12 px
WEIGHT	600
CASE	Uppercase, 0.14em
COLOR	--accent (#1A5A8E)

CHART TITLE

Copper exporters cluster by complexity, not size.

Headline of the chart. Always ends in a period — it reads as a finding, not a label.

FAMILY	Source Serif 4
SIZE	14 px
WEIGHT	500
LEADING	1.25
TRACKING	-0.005em
COLOR	--ink

CHART SUBTITLE

GDP per capita vs. economic complexity, 2022.

Gives the chart its units, period, and unit of analysis. Omit when redundant with the title.

FAMILY	Inter
SIZE	12 px
WEIGHT	400
LEADING	1.4
COLOR	--ink-3

AXIS LABEL

% real GDP growth

Names the dimension and its units. One short sentence-case phrase — never all caps.

FAMILY	Inter
SIZE	12 px
WEIGHT	500
COLOR	--ink-2 (#2C2823)
PLACEMENT	X: centered below ticks. Y: rotated -90°, centered left of ticks.

AXIS TICK LABEL

Numeric values along each axis. Always tabular figures so digits align across ticks.

0 25 50 75 100

FAMILY	Inter
SIZE	12 px
WEIGHT	400
COLOR	--ink-2 (#2C2823)
FIGURES	font-variant-numeric: tabular-nums

SERIES LABEL

Series 1 Series 2

*Direct labels at the end of each line, or a small legend below. Text always uses the **darker version** of the series color (--c-N-dark), never the main tone — this is required to clear WCAG AA contrast against the white background.*

FAMILY	Inter
SIZE	12 px
WEIGHT	600
COLOR	Dark category color (e.g. --c-1-dark)

CHART SOURCE

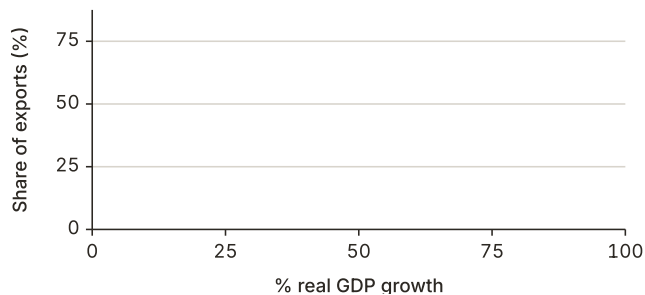
Source: Growth Lab analysis of UN Comtrade.

Always required. One line below the chart. Italic serif distinguishes provenance from the data narrative.

FAMILY	Source Serif 4 italic
SIZE	12 px
LEADING	1.45
COLOR	--ink-2

Axes, ticks, gridlines

Axes carry the frame of reference; they should be present but quiet. Use a single 1-px line for each axis, short outward ticks, and pale gridlines only where they help the reader estimate values.



A bare axis: 1-px ink line, 4-px outward ticks, pale gridlines, sentence-case labels.

AXIS LINE	1 px solid --axis (#2C2823)
TICK MARK	1 px, 4 px long, outward, --axis
GRIDLINE	1 px solid --gridline (#D8D4CC)
TICK LABEL	Inter 12 px / 400 / --ink- 2 / tabular-nums
AXIS LABEL	Inter 12 px / 500 / --ink- 2
TICK OFFSET	Tick label sits 6 px outside the axis
LABEL OFFSET	Measure axis-label distance from the <i>start</i> of the tick label, not from the axis line. On Y that means 20 px to the left of the leftmost character of the widest tick label — otherwise wide ticks like “250” collide with the rotated axis label. Same 20 px offset on X, below the tick baseline.
YEAR AXIS	When the X axis is just years, omit the axis label — the tick labels already say what the dimension is
ZERO BASELINE	Stroke ink, never gridline weight

Marks — circles, lines, areas

Every category color comes as a fill / stroke pair. **Scatter circles are the only mark that uses 0.8 fill opacity** — every other filled mark (bars, areas, treemap tiles, choropleth polygons) renders at full opacity. The stroke is always a darker tone of the same hue at full opacity.



Scatter circle — the only mark with reduced fill opacity. Lets overlapping points darken visibly to signal density.

SHAPE	Circle, default radius 5–7 px
FILL	Category color (e.g. --c-1) at fill-opacity: 0.8
STROKE	Matching darker tone (e.g. --c-1-dark) at stroke-opacity: 0.8 — same as the fill, so overlapping points darken together
STROKE WIDTH	1 px
SCOPE	Scatter only — do not apply 0.8 opacity to bars, areas, treemap, or maps



Bar / area / treemap fills — main palette color at full opacity, no stroke.

FILL	Main category color (e.g. --c-1) at full opacity
STROKE	None — tiles abut directly
APPLIES TO	Stacked bars, area bands, treemap tiles



Line marks use the main palette color at full opacity (no fill, no darker tone).

STROKE	Main category color (e.g. --c-1)
WIDTH	2 px standard / 2.4 px for the highlighted focus series
JOIN	round; stroke-linejoin: round
STYLE	Solid only. Avoid dashed lines except for projections.

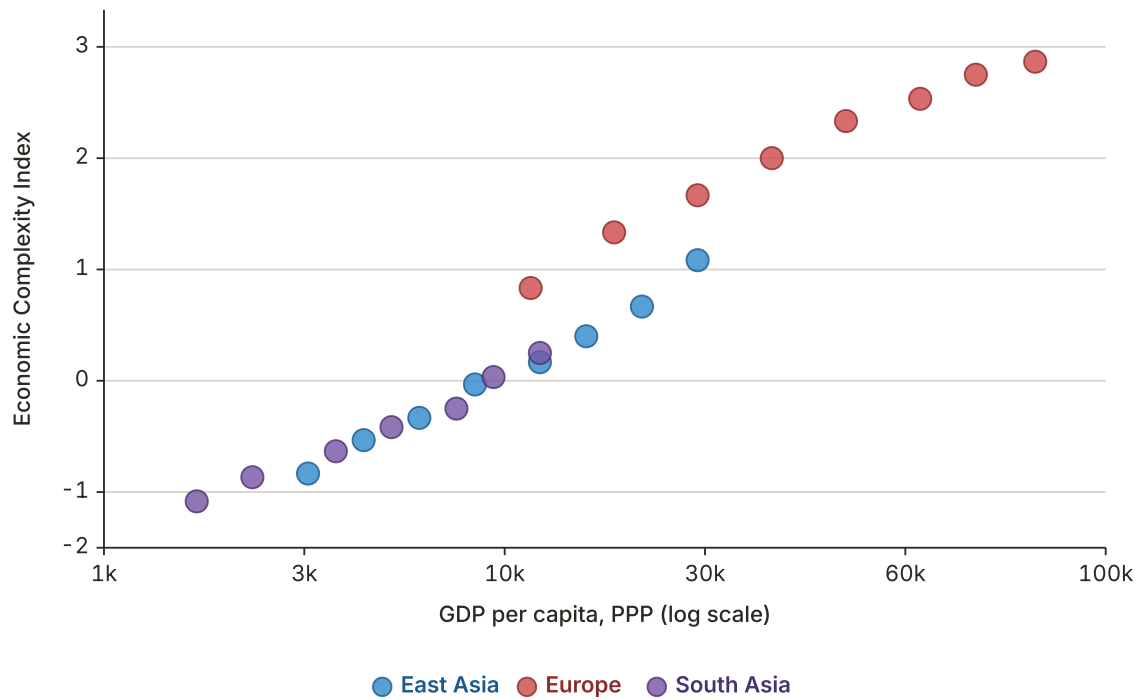
Scatter plot

Use a scatter plot when the story is the relationship between two continuous variables. Each point is a category-fill circle at 0.8 opacity with a darker stroke at the same 0.8 opacity. Overlap is desirable — it shows clustering.

FIGURE 1

Economic complexity tracks income, with notable outliers.

GDP per capita (PPP, log scale) vs. Economic Complexity Index, by region, 2022.



Source: Growth Lab analysis of Atlas of Economic Complexity, 2022.

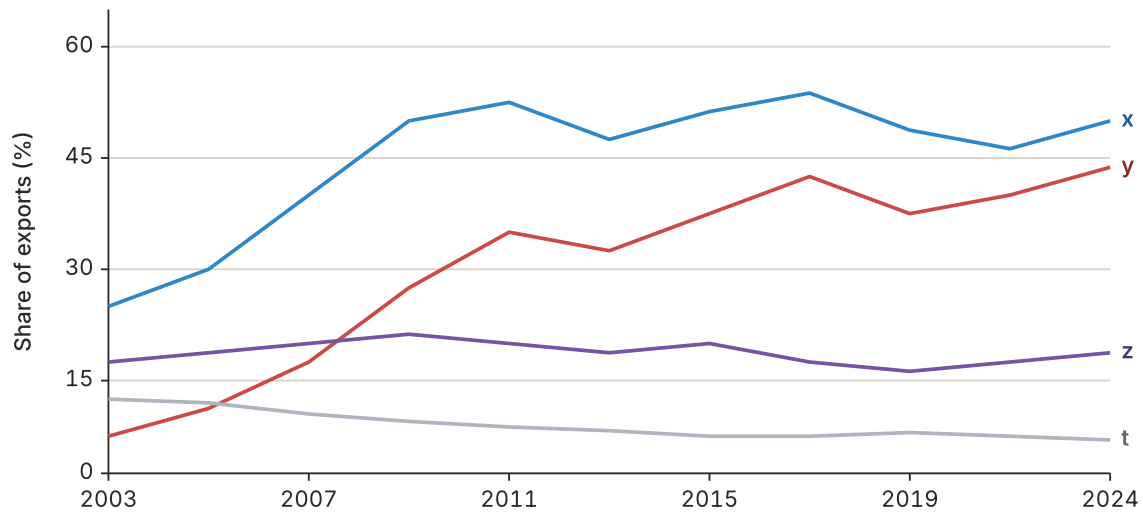
Line chart

Use a line chart when the story is change over a continuous variable, usually time. Up to four series; beyond that, mute the supporting lines in `--c-muted` and let the focus series carry the finding.

FIGURE 2

Copper and coal carried the Mongolian boom.

Share of merchandise exports by category, 2003–2024, percent.



Source: Growth Lab analysis of UN Comtrade. Categories: Copper (HS 2603, 7402); Coal (HS 2701, 2702).

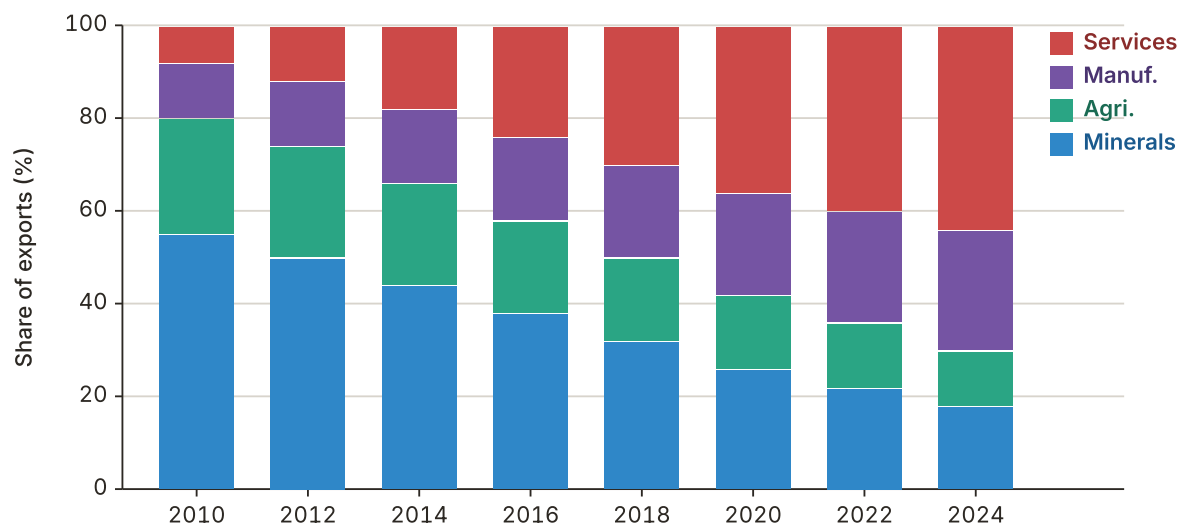
Stacked chart

Stacked bars (or areas) show how a total composes. Use up to six categories, ordered from largest mean share at the bottom upward. Fills are the main palette color at full opacity, no stroke. A **1 px gap** separates each segment from the one above — this gives the boundary between categories a clean edge and helps readers with low color discrimination tell adjacent segments apart.

FIGURE 3

Services took over the export basket.

Export composition by sector, 2010–2024, share of total exports.



Source: Growth Lab analysis of national statistics.

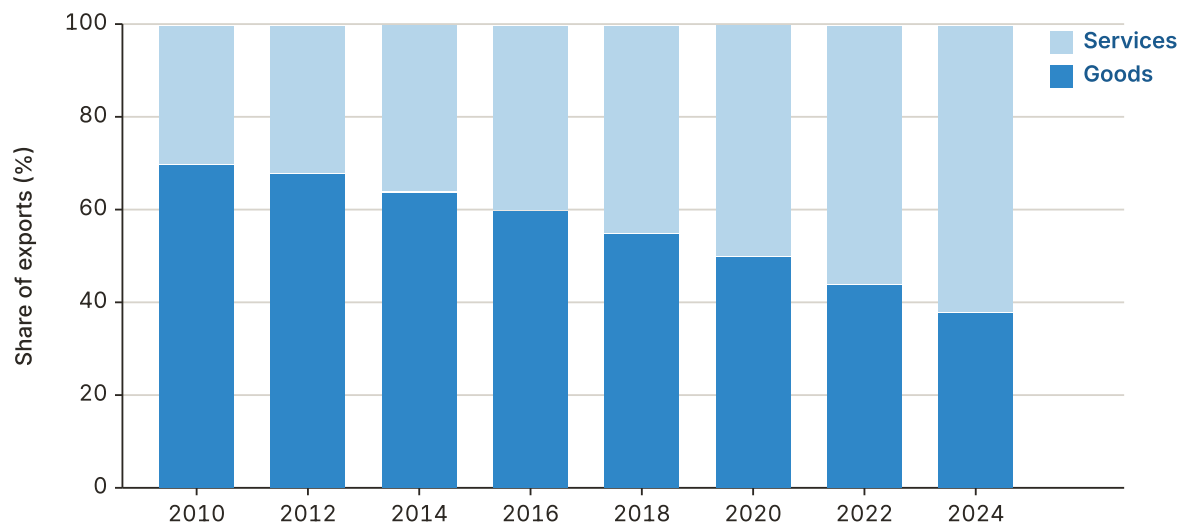
Two or three tone option

Whenever possible, try to limit the use of different colors and use the different tones of the same color instead. You can also do this for categorical data: when two or three categories share a common parent — goods vs. services, low/medium/high tier, primary/intermediate/final — encode them with a single hue at its light, main, and dark tones rather than reaching for three unrelated colors. The shared hue keeps the bar reading as one total; the lightness difference carries the split. The 1 px gap between segments still applies.

FIGURE 3B

Services overtook goods in the export basket.

Goods vs. services share of total exports, 2010–2024, percent.



Source: Growth Lab analysis of national statistics.

Two tones, c-1 main and c-1-light. The same pattern works with any other categorical hue — --c-2 with --c-2-light, --c-4 with --c-4-light, and so on. Reach for it when the chart sits inside a section where that hue already carries narrative weight.

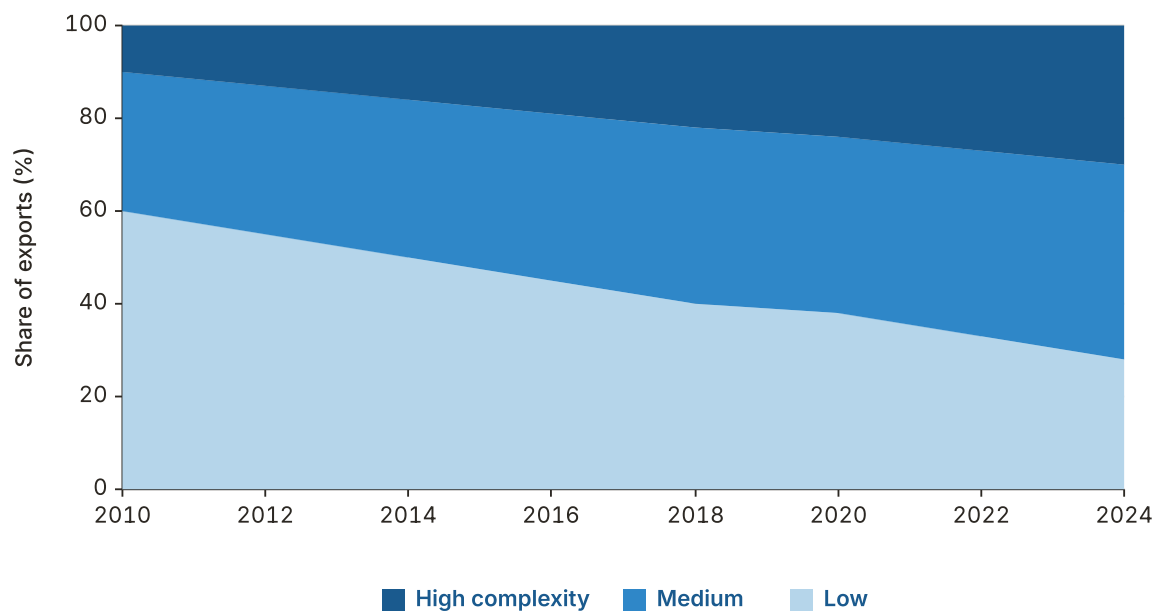
Two or three tone option

When a third category fits within the same parent concept, extend the same idea to all three tones of the hue: `--c-1-light`, `--c-1`, and `--c-1-dark`. Stacked-area form is the natural reading: the layers flow continuously across time, and the lightness ramp itself carries the ordering — lightest for the lowest tier at the bottom, darkest for the highest at the top. This is the one place outside a scatter plot where the dark tone is allowed.

FIGURE 3C

High-complexity exports grew share over the decade.

Composition of merchandise exports by complexity tier, 2010–2024, percent.



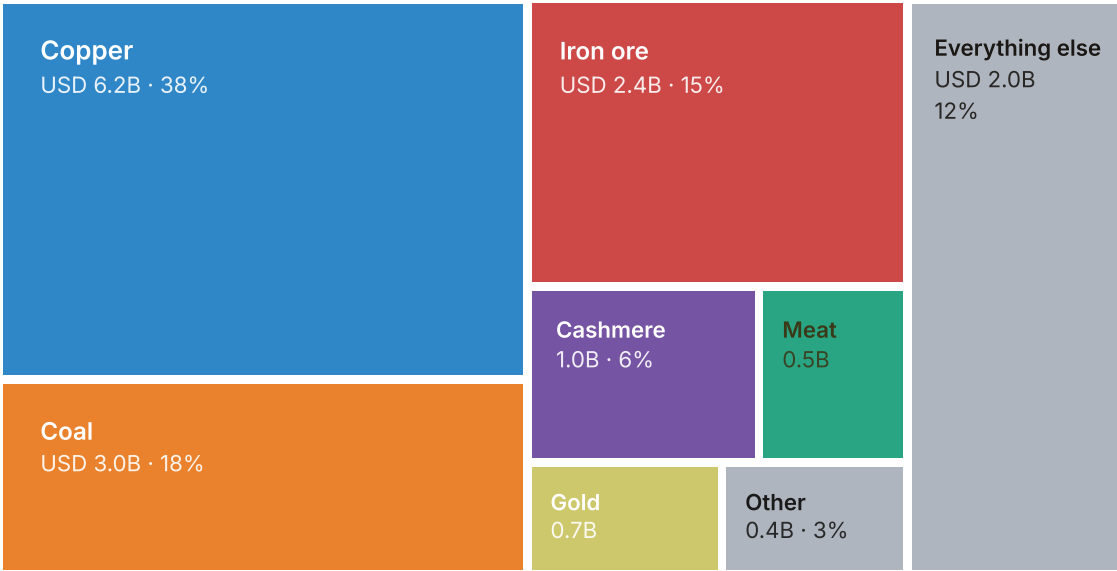
Source: Growth Lab analysis of UN Comtrade. Complexity tiers derived from PCI percentile.

Three tones, `c-1-light` / `c-1` / `c-1-dark`. The dark tone is allowed here because all three bands belong to the same parent variable; the reader sees one ordered hue, not a scatter-circle stroke. The same approach scales to any hue (`--c-2-light` / `--c-2` / `--c-2-dark`, etc.). Area bands sit edge-to-edge with no stroke and no gap — the lightness step is what separates them.

Tree map

Use a tree map when the story is the composition of a whole across many categories with very uneven shares. Each tile uses category fill at 0.8 opacity and the matching darker stroke at full opacity along its edge.

FIGURE 4
A handful of products dominate the export basket.
Tiles sized by 2024 export value, USD billions.

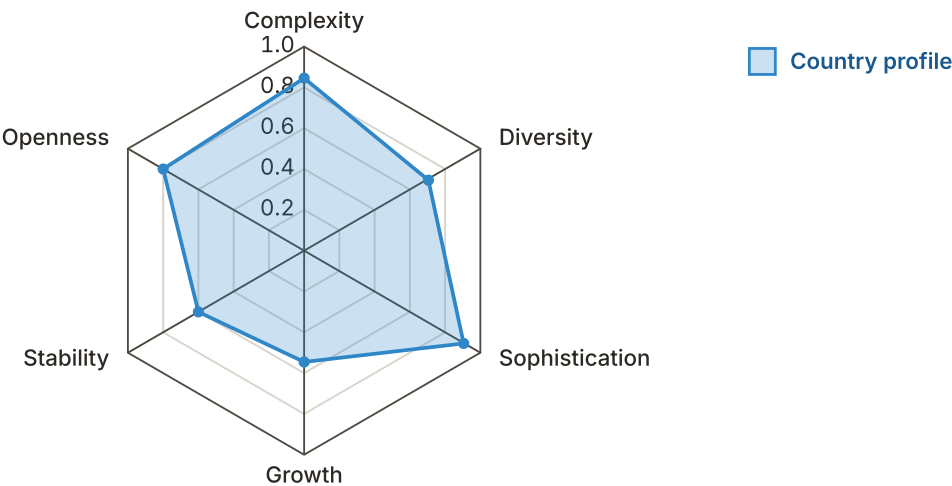


Source: Growth Lab analysis of UN Comtrade, 2024.

Radar chart

Use a radar chart to profile a single entity across four to eight dimensions on a shared normalized scale (0–1, 0–100, or any common index). The polygon’s shape carries the story; its area is a quick read of overall standing. The fill is `--c-1 (#2F87C8)` at **fill-opacity: 0.25** so the gridlines and axis labels remain readable through it; the outline uses the same hue at full opacity, 2 px.

FIGURE 5
A balanced complexity profile, weakest on growth volatility.
Six-dimension capability index, normalized 0–1, latest year.



Source: Growth Lab analysis. Dimensions normalized to the cross-country max in each axis.

Radar at a glance — the same construction rule as scatter (fill at reduced opacity, full-opacity stroke), but tuned for the larger filled area.

SERIES FILL	--c-1 (#2F87C8) at fill-opacity: 0.25
SERIES STROKE	--c-1 at full opacity, 2 px, stroke-linejoin: round
VERTEX DOT	3 px radius, filled --c-1, no stroke
GRID RINGS	1 px --gridline; outermost ring in --ink-3
AXIS LINES	1 px --ink-3 from center to each vertex of the outer ring
AXIS LABELS	Inter 12 px / 500 / --ink-2, set just outside the outer ring
SCALE TICKS	Inter 12 px / 400 / --ink-2, placed along the top

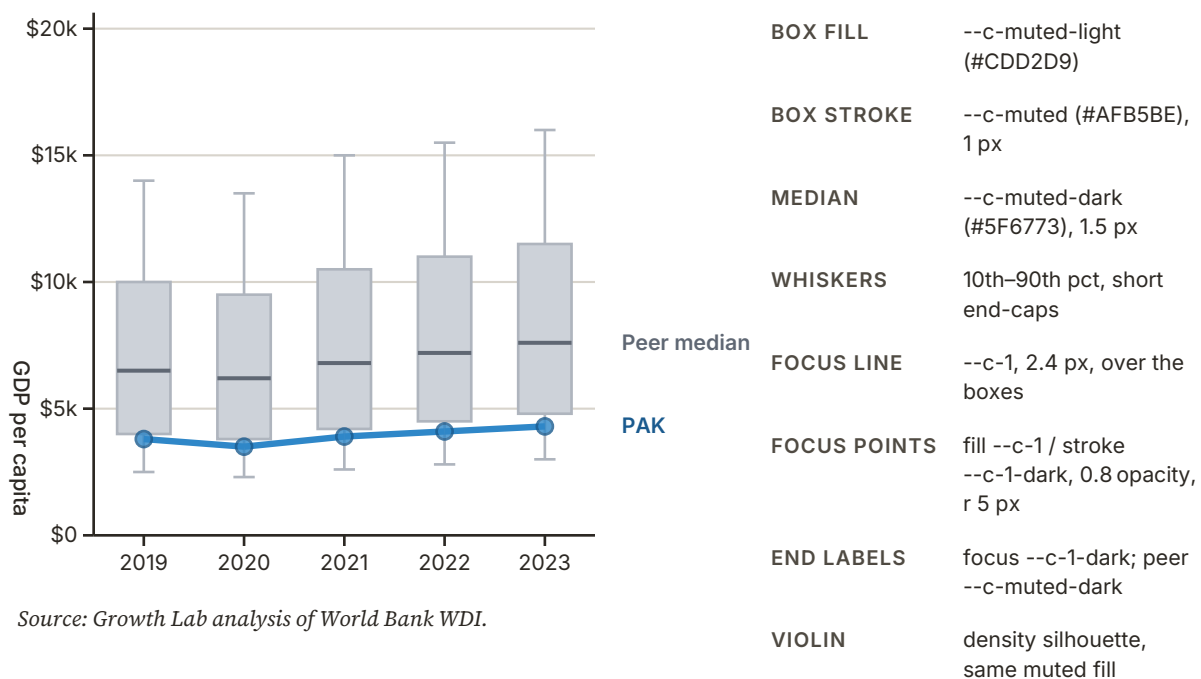
Boxplot & violin

Use a box plot when the story is the spread of a distribution — how a value ranges across a group — not a single point or a trend. The box spans the interquartile range; the whiskers reach the 10th and 90th percentiles. Keep the supporting distribution muted and draw one focus entity as a c-1 line on top, so the reader tracks it against the spread. A violin is the same construction with a density silhouette in place of the box.

FIGURE 6

Pakistan sits in the lower half of the regional income distribution.

GDP per capita by year across regional peers, 2019–2023, USD.



**SECOND
SERIES**

axis only

If overlaying a
comparison, paint it in --
c-muted at the same
opacity and stack it *under*
the focus series

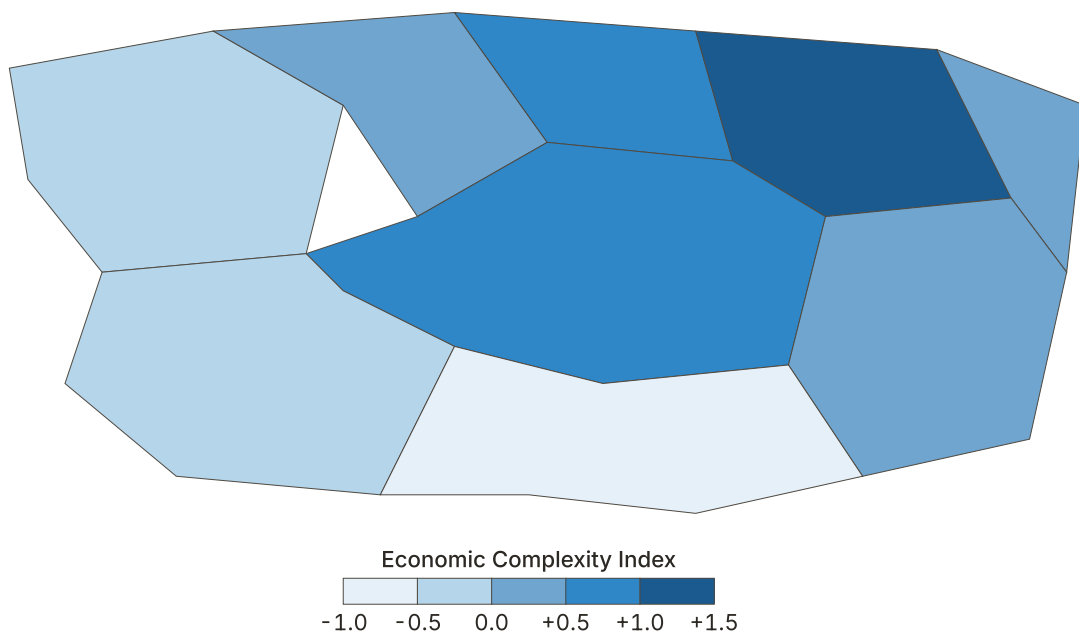
Geomap — sequential

Use a sequential choropleth when the encoded value runs from low to high without a natural midpoint — population, GDP, complexity. Five steps on a single blue hue; darker means higher. Polygons render at full opacity; a 0.5-px ink-3 stroke separates regions.

FIGURE 7

Complexity is concentrated in the central provinces.

Economic Complexity Index by province, 2022.



Source: Growth Lab analysis. Provinces stylized for illustration.

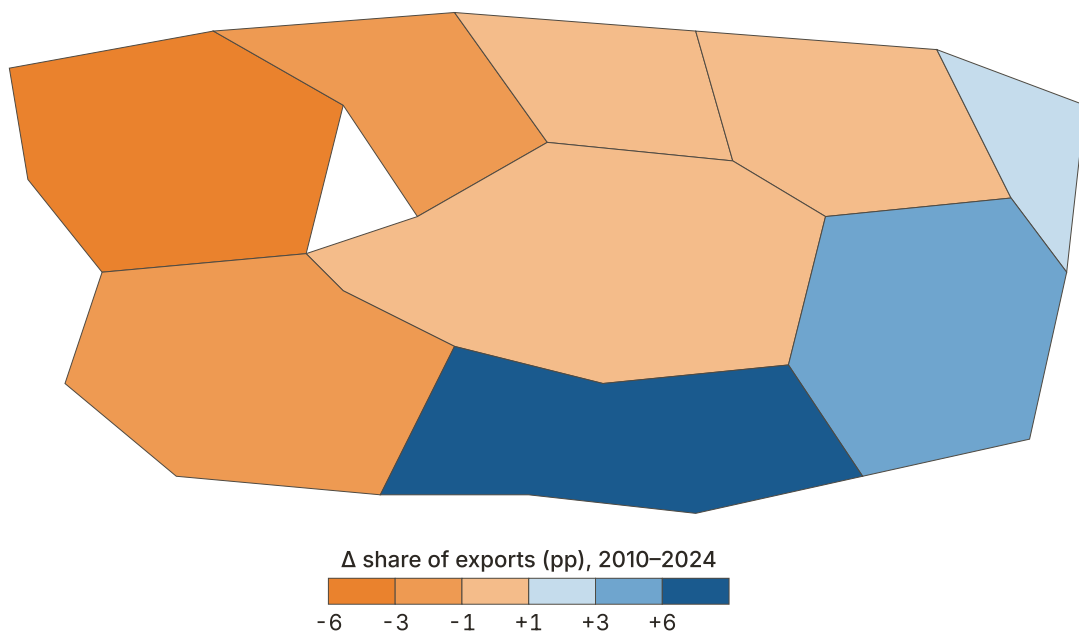
Geomap — diverging

Use a diverging choropleth when the value has a meaningful midpoint — change vs. baseline, gain vs. loss, above vs. below the national average. Red for the negative tail, blue for the positive tail, near-white at the midpoint. Polygons full opacity; ink-3 stroke at 0.5 px.

FIGURE 8

Coastal provinces gained share; the interior lost it.

Change in share of national exports, 2010 vs. 2024, percentage points.



Source: Growth Lab analysis of national export records.

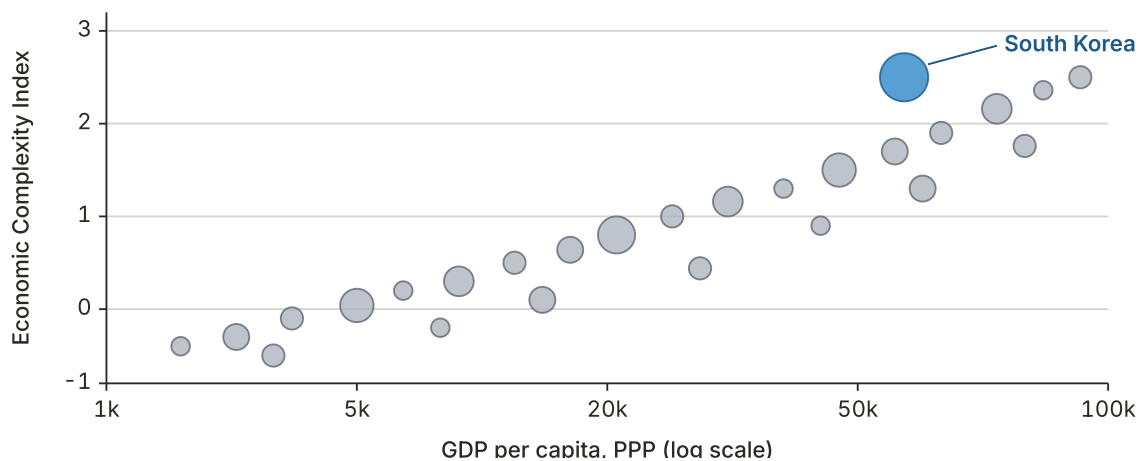
Implementing the pop-up effect

Color should only be used when it is necessary. When you have categorical data, you do not have to color every category differently. Always ask first whether the story can be told by a **pop-up effect** instead — paint the supporting data in `--c-muted` (`#AFB5BE`) and reserve a saturated hue (usually `--c-1`, sometimes `--c-1` and `--c-2` together) for the one or two series the reader actually needs to track. The muted layer carries the trend; the highlight carries the finding.

FIGURE 9

South Korea climbed the complexity ladder faster than its peers.

GDP per capita vs. Economic Complexity Index, bubble size = export value, 2022.

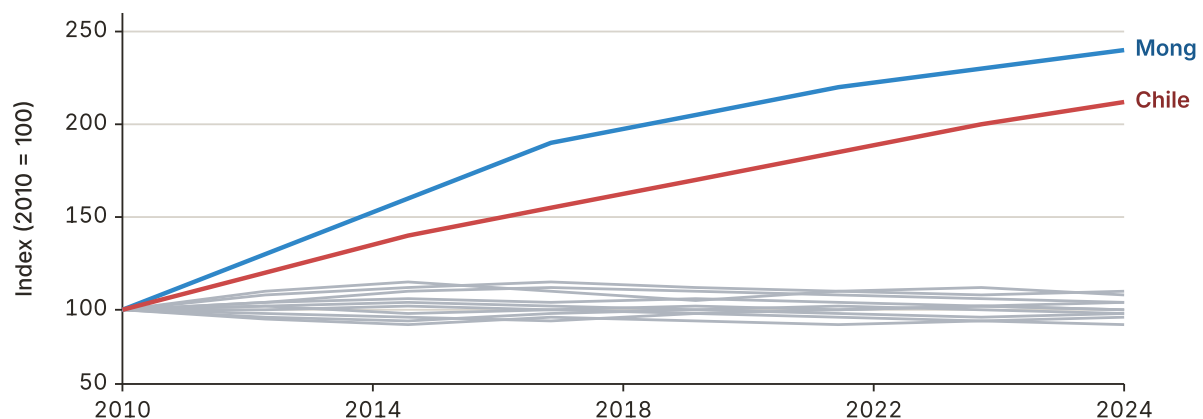


Source: Growth Lab analysis of *Atlas of Economic Complexity*, 2022.

FIGURE 10

Mongolia and Chile broke from the pack on copper exports.

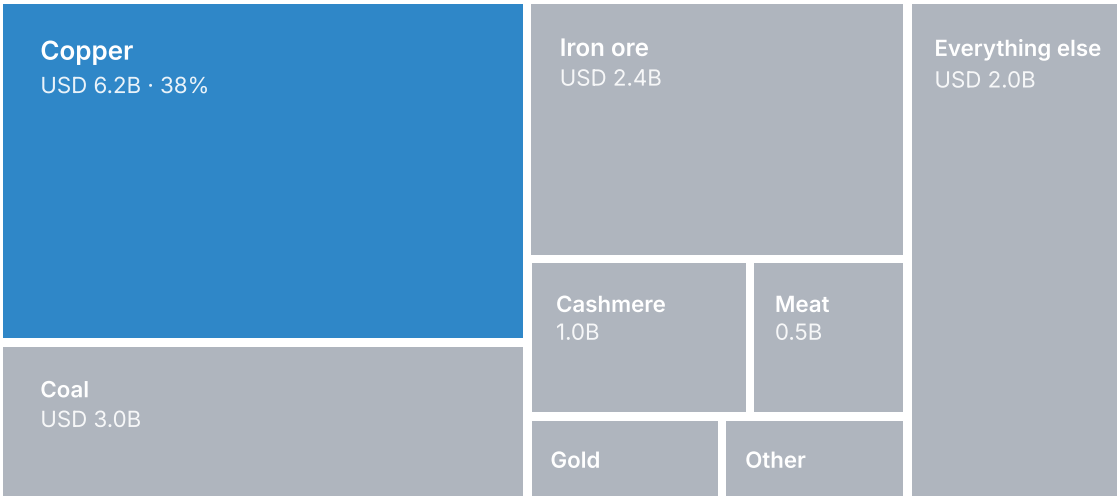
Index of copper export value (2010 = 100), twelve mineral economies, 2010–2024.



Source: Growth Lab analysis of UN Comtrade. HS 2603, 7402.

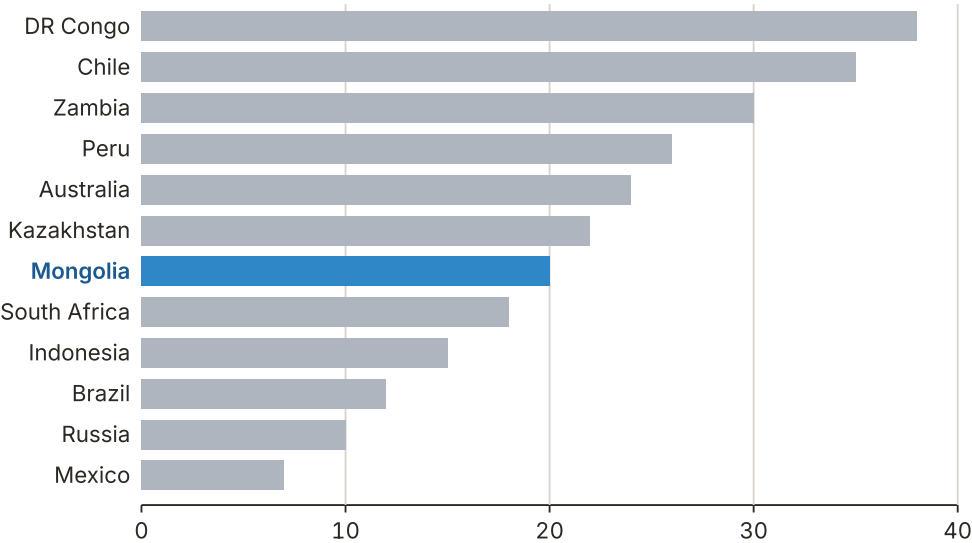
Implementing the pop-up effect

FIGURE 11
Copper alone accounts for nearly 40% of exports.
Tiles sized by 2024 export value; only the focus category is colored.



Source: Growth Lab analysis of UN Comtrade, 2024.

FIGURE 12
Mongolia ranks seventh among emerging mineral exporters.
Mineral export share of GDP, twelve economies, 2024.



Source: Growth Lab analysis of national accounts, 2024.

The pattern in every figure above: every supporting mark in `--c-muted` (`#AFB5BE`), one or two focus marks in `--c-1` (and optionally `--c-2` for a secondary), labels in matching focus color. Before reaching for a categorical palette, ask whether a pop-up does the work.

Decision rules

Before adding a new chart style, work through this list.

1. **Seven-color palette including a muted color.** Colors are added in order — `--c-1` first, then `--c-2`, `--c-3`, and so on. All six categorical colors should only be used if there is absolute necessity. If a visualization needs more than six, the representation mode should be re-thought.
2. **The darker tone is for strokes and all text associated with the visualization.** Strokes around scatter circles use the dark version of the fill colour; every direct label, legend entry, callout, and annotation also uses the dark tone. The main tones do all the fill work — only stroke and text reach for the dark variant. This is required to clear WCAG AA contrast against the white background.
3. **Overlapping encodings get a 0.8 opacity.** Whenever marks can sit on top of each other — scatter circles, area bands, overlaid polygons — apply `fill-opacity: 0.8` and `stroke-opacity: 0.8` together. The shared 0.8 lets overlapping marks darken visibly into the underlying layer, signalling density, while non-overlapping marks still read as the full main tone. Single-layer marks (bars, treemap tiles, choropleth polygons) stay at full opacity.
4. **Axis line and ticks are 1 px `--axis (#2C2823)`.** Ticks are 4 px long, outward, never inward.
5. **Gridlines are 1 px `--gridline (#D8D4CC)`.** Use them only where the reader needs to estimate a value off the axis; never both X and Y unless the chart is dense.
6. **Axis labels: Inter 12 px / 500 / `--ink-2`.** Tick labels: Inter 12 px / 400 / `--ink-2` with `tabular-nums`.
7. **Color encodes meaning, not decoration.** If a color is not earning its keep — a category, a threshold, a highlight — remove it.
8. **Whenever possible, implement the pop-up effect.** Mute the supporting data in `--c-muted` and reserve a saturated hue for the one or two series the reader actually needs to track. See Section 13 for worked examples.
9. **Sequential ramp for ordered values; diverging ramp only with a meaningful midpoint.** Never use diverging for purely positive scales.
10. **Chart title ends with a period.** It reads as a finding statement. Subtitle does not.
11. **No monospace.** All numerals use Inter with `font-variant-numeric: tabular-nums`.